SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	2 July 2026
Team ID	xxxxxx
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	*HTML, CSS, JavaScript, Bootstrap*	*Provides a responsive and user-friendly interface for entering rainfall and weather data and displaying flood prediction results.*

2	**Backend / Server-Side**	*Python, Flask*	*Flask is lightweight and integrates easily with machine learning models. Python supports efficient data processing and prediction.*
3	**Database / Data Storage**	*MySQL*	*Stores historical flood records, rainfall data, user inputs, and prediction results securely and efficiently*
4	**Cloud / Hosting / Deployment**	*Render*	*Enables easy deployment of the web application with minimal configuration and supports future scalability.*
5	**Version Control & CI/CD**	*GitHub*	*Used for source code management, collaboration among team members, and maintaining version history.*

6	**Third-Party APIs / Other Tools**	*OpenWeatherMap API, Scikit-learn, Pandas, NumPy, Matplotlib*	*OpenWeatherMap API provides real-time weather data. Scikit-learn is used for machine learning algorithms, Pandas and NumPy for data processing, and Matplotlib for visualization of flood trends and prediction results*